FACULTY OF COMPUTING
UTM Johor Bahru

SECD2613: System Analysis and Design

2025/2026-2

ANALYSIS AND DESIGN
FlexHub AI Marketplace (FlexServe)

Phase 3

Section : 01
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1. Overview of the Project

Service providers ranging from makeup artists and consultants to private tutors are rapidly expanding. However, the digital platforms for supporting this expertise often rely on rigid, traditional booking models. Existing platforms fail to help service providers to better manage their time and connect users to the services they need. FlexHub (FlexServe) aims to solve this issue by implementing an AI time management system and better help users match a service provider based on their time and need. This platform will facilitate seamless connections between clients searching for specific expertise and freelancers offering their services by the hour or by the session.

2. Problem Statement

Most commonly, clients seeking professional services must rely on a fragmented ecosystem, as each service provider has their own isolated social media pages or static booking websites. The lack of a centralized website hosting huge ranges of different services makes it difficult for a client to discover experts who match their specific needs and schedule constraints. For example, if someone wants to hire a landscaper to help decorate their garden and rely on the few social media pages they follow, they will have very limited choices to choose from causing they might be unable to find the most suitable landscaper or have a hard time scheduling appointments with the landscaper they might choose.

The few centralized websites or platforms that exist are, however, too static. Consequently, both professional experts and the client using the platforms struggle with inefficient scheduling systems. Static calendars frequently lead to double bookings, and also unexpected scheduling conflicts that happen because their system is unable to factor in buffer times between appointments. These legacy systems fail to account for the required buffer times, such as considering the travel between client locations, or a human requirement of the preparation and cleanup between sessions. Furthermore, existing platforms strictly categorize users as either service providers or clients only, so there is no flexibility for individuals who wish to operate as service providers and become consumers for other type of services officered within the same ecosystem.

3. Proposed Solutions

We propose developing FlexServe, a sub-system of the overarching FlexHub AI Marketplace, designed specifically to manage professional experts acting as freelancers. This centralized platform will empower users to seamlessly discover, match, and book experts based on availability and preferences.

To address the limitations of static platforms, the system will incorporate one core AI integrations:

- **AI Smart Scheduler:** To eliminate booking conflicts, the system will utilize an intelligent time-slot management algorithm. This AI feature will dynamically calculate and enforce "buffer times" between bookings, automatically accounting for estimated travel time or session cleanup, ensuring a realistic and stress-free schedule for the provider.

Additionally, the platform will analyze user preferences, past interactions, and service requirements to recommend the most suitable experts, ensuring a high-quality match between the client's needs and the provider's skills. The system will also feature Role Management, allowing a single user account to seamlessly toggle between being a Client and Service Provider.

4. Current Business Process/Workflow

4.1 Current Business Process

Based on AS-IS analysis

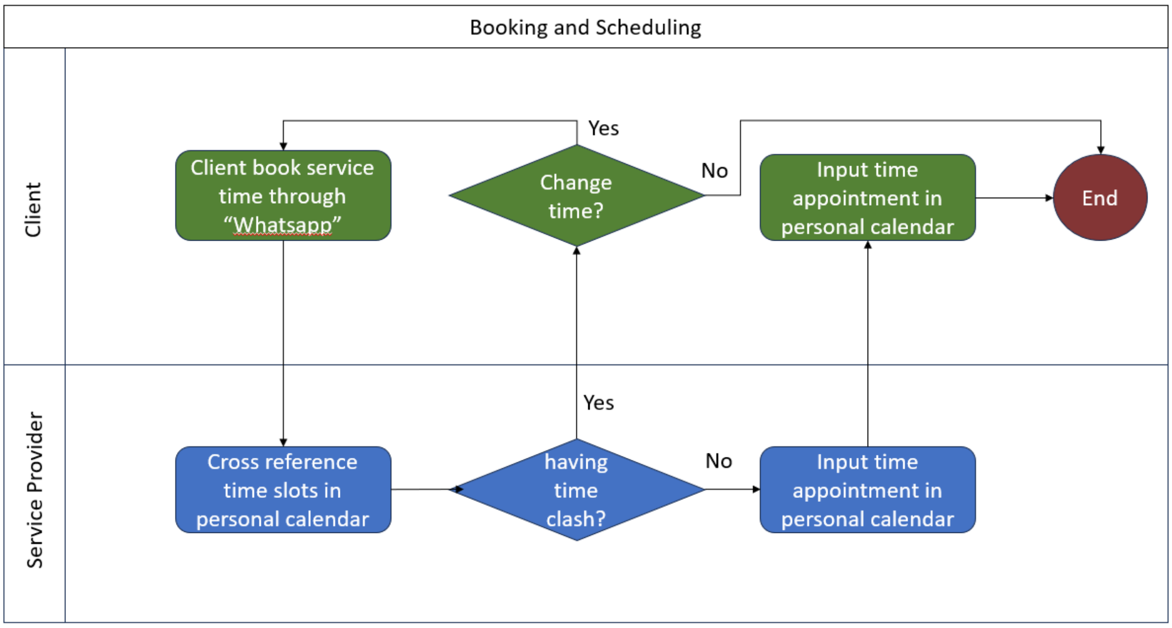

Figure 4.1.1 Booking and scheduling flowchart

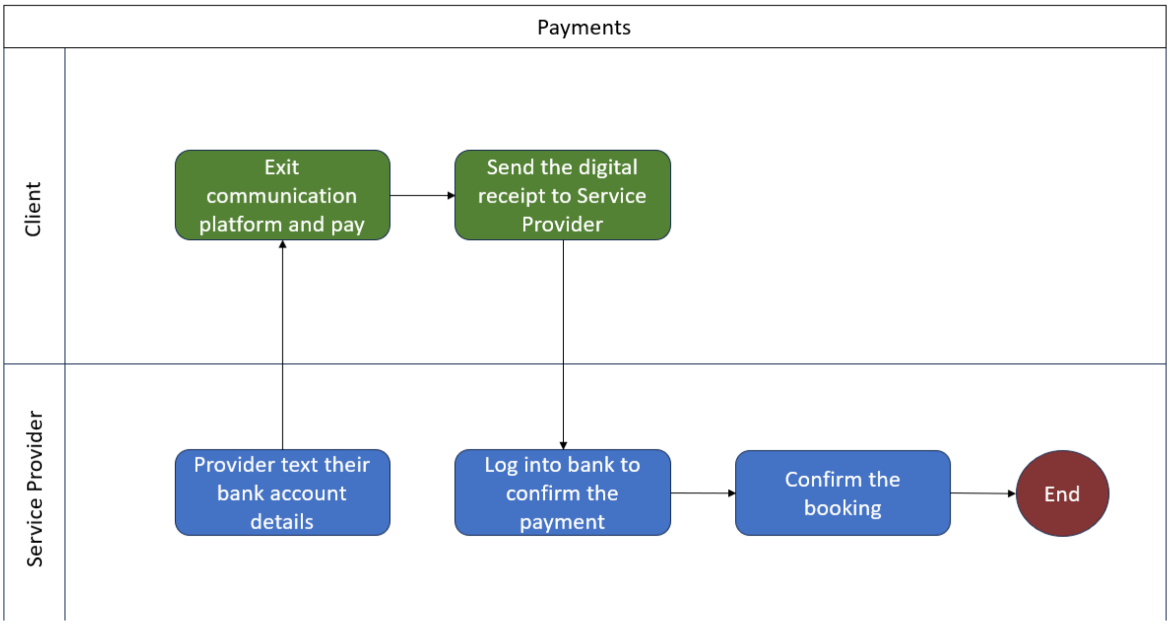

Figure 4.1.2 Payments flowchart

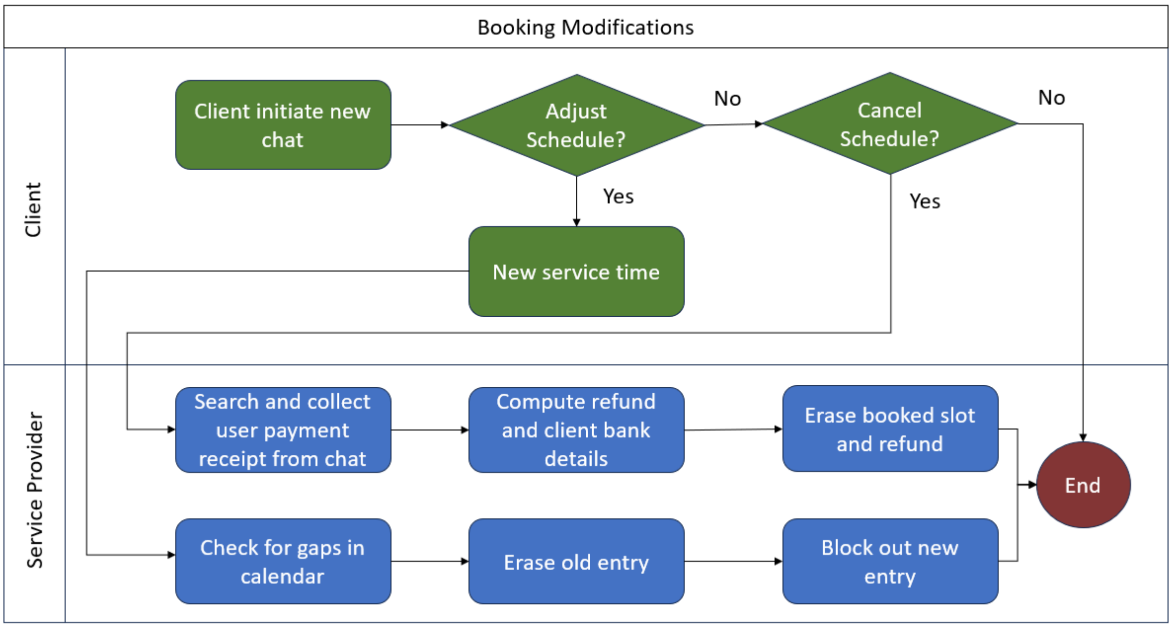

Figure 4.1.3 Booking modifications, cancellation and refunds flowchart

5. Logical DFD (AS-IS)

5.1 Logical DFD AS-IS system

Context Diagram:

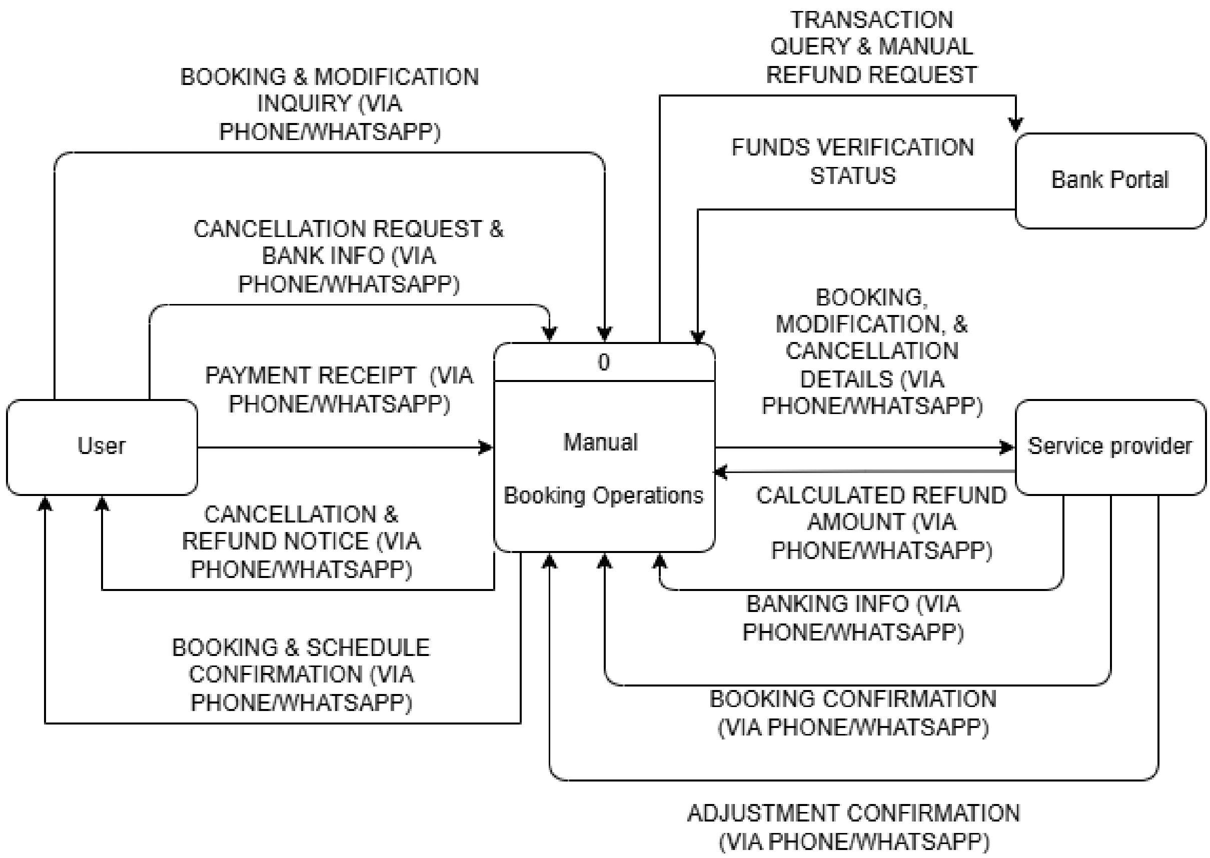

Figure 5.1.1 Context diagram of existing system

Diagram 0:

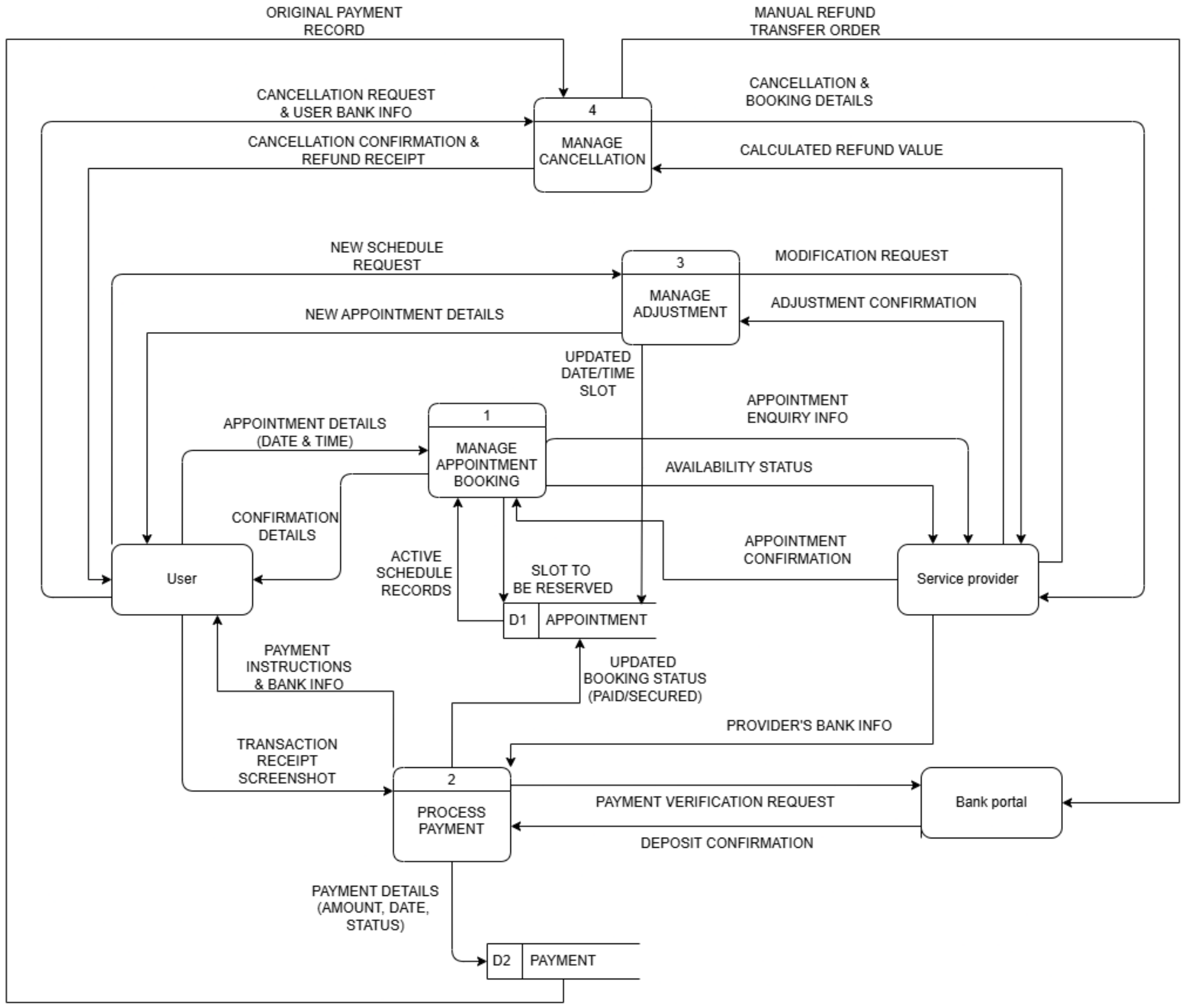

Figure 5.1.2 Diagram 0 of existing system

Child diagram:

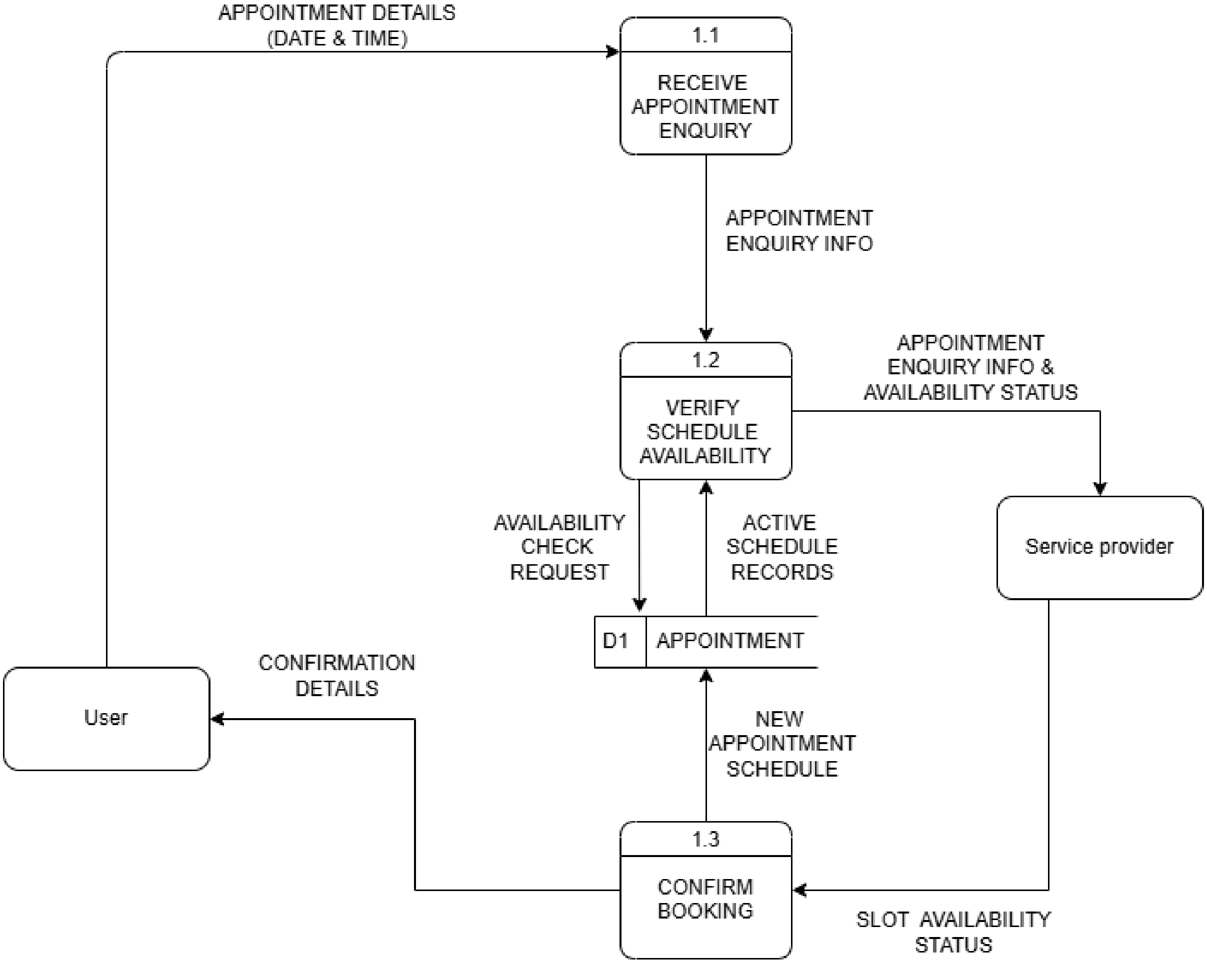

Figure 5.1.3 Child diagram of Process 1

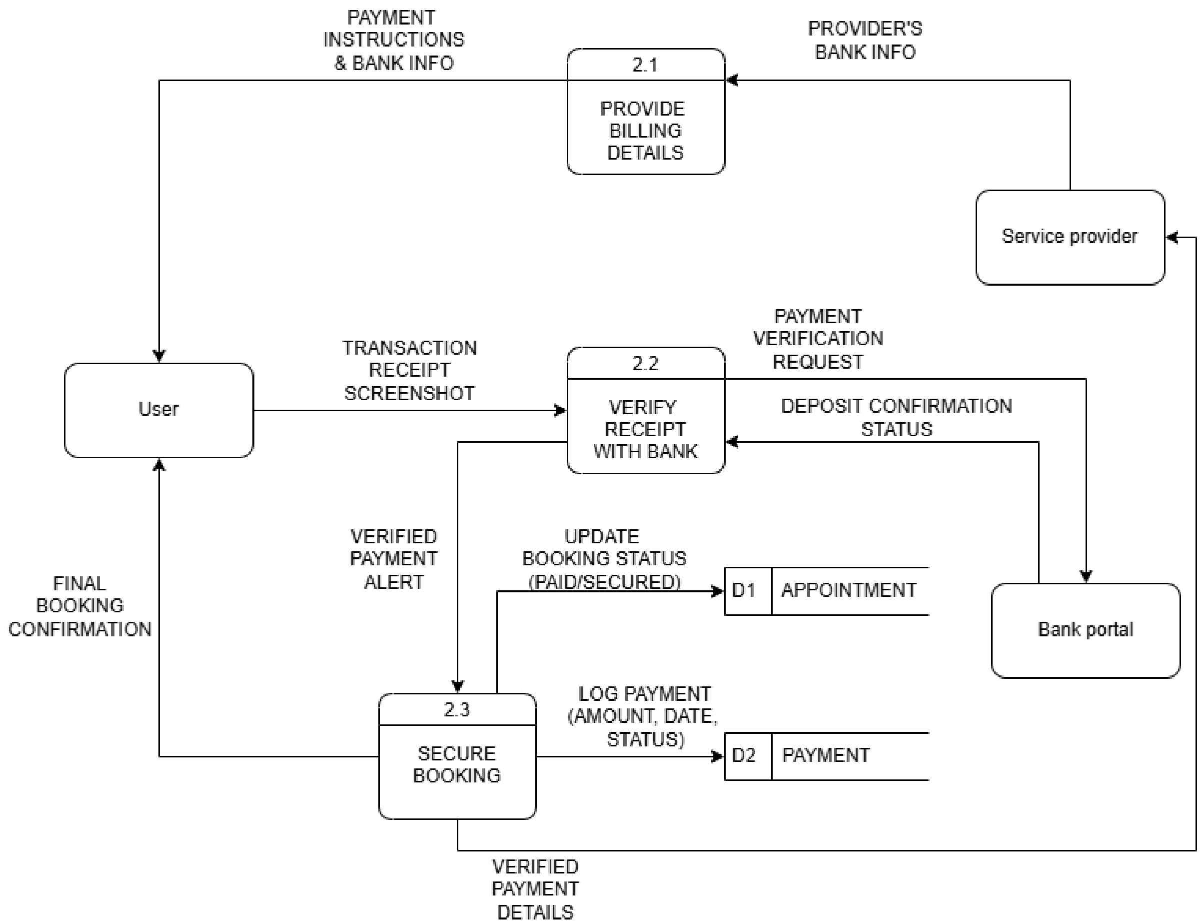

Figure 5.1.4 Child diagram of Process 2

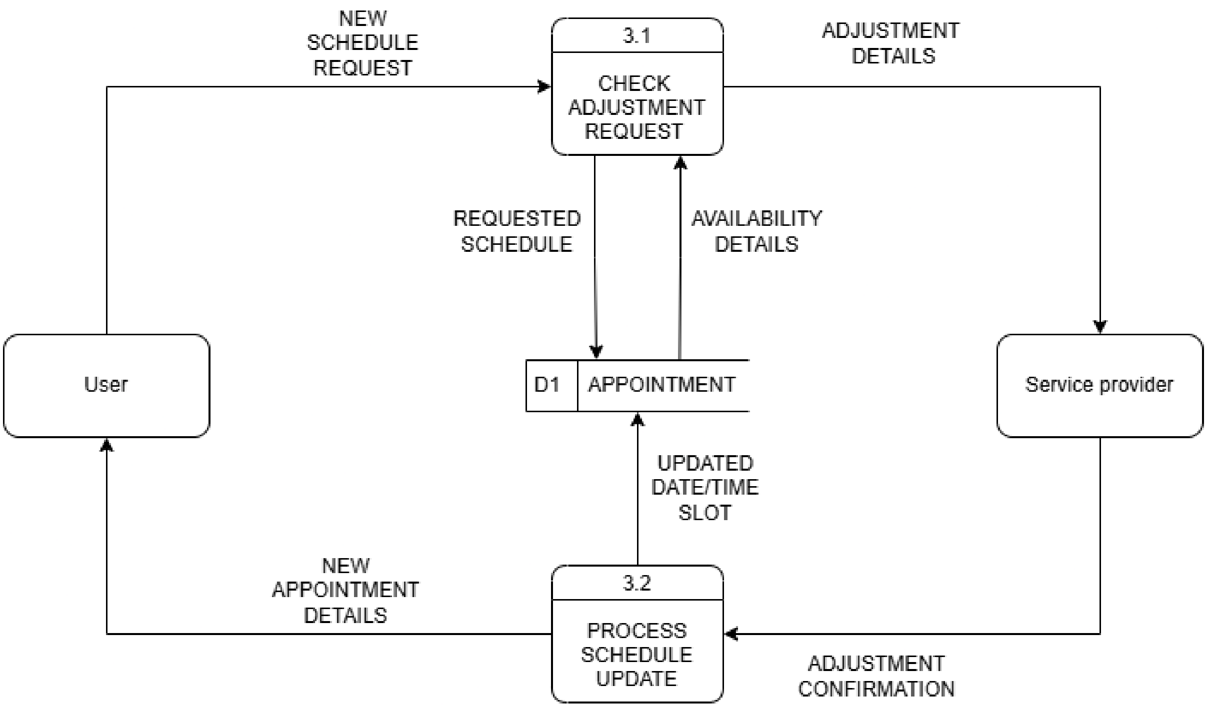

Figure 5.1.5 Child diagram of Process 3

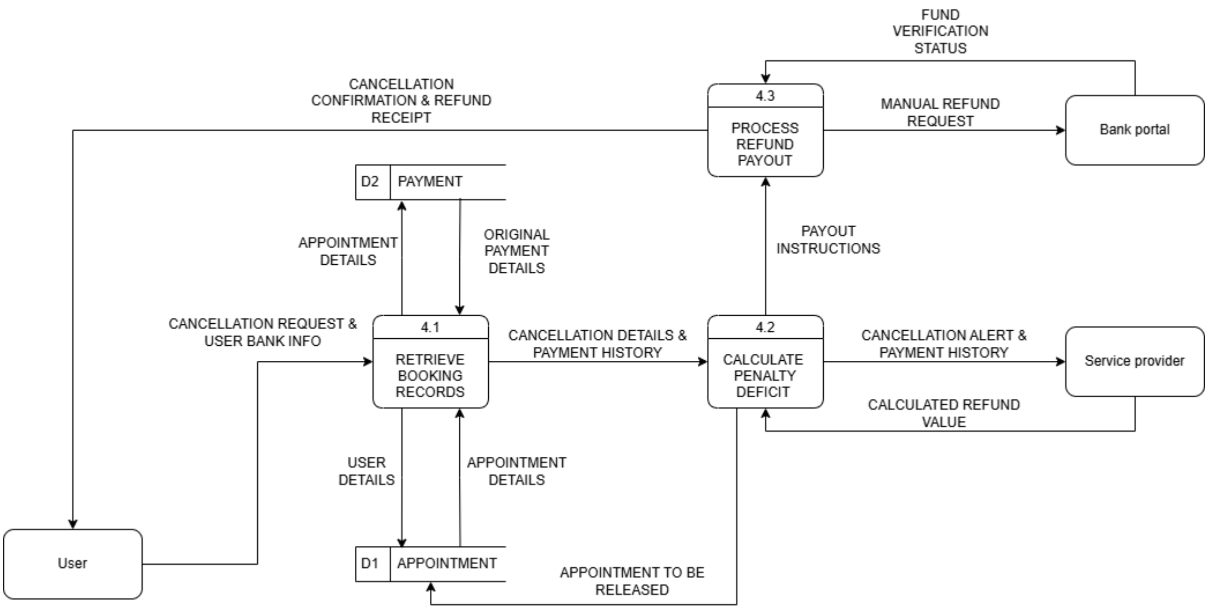

Figure 5.1.6 Child diagram of Process 4

6. System Analysis and Specification

6.1 Logical DFD TO-BE system

Context Diagram:

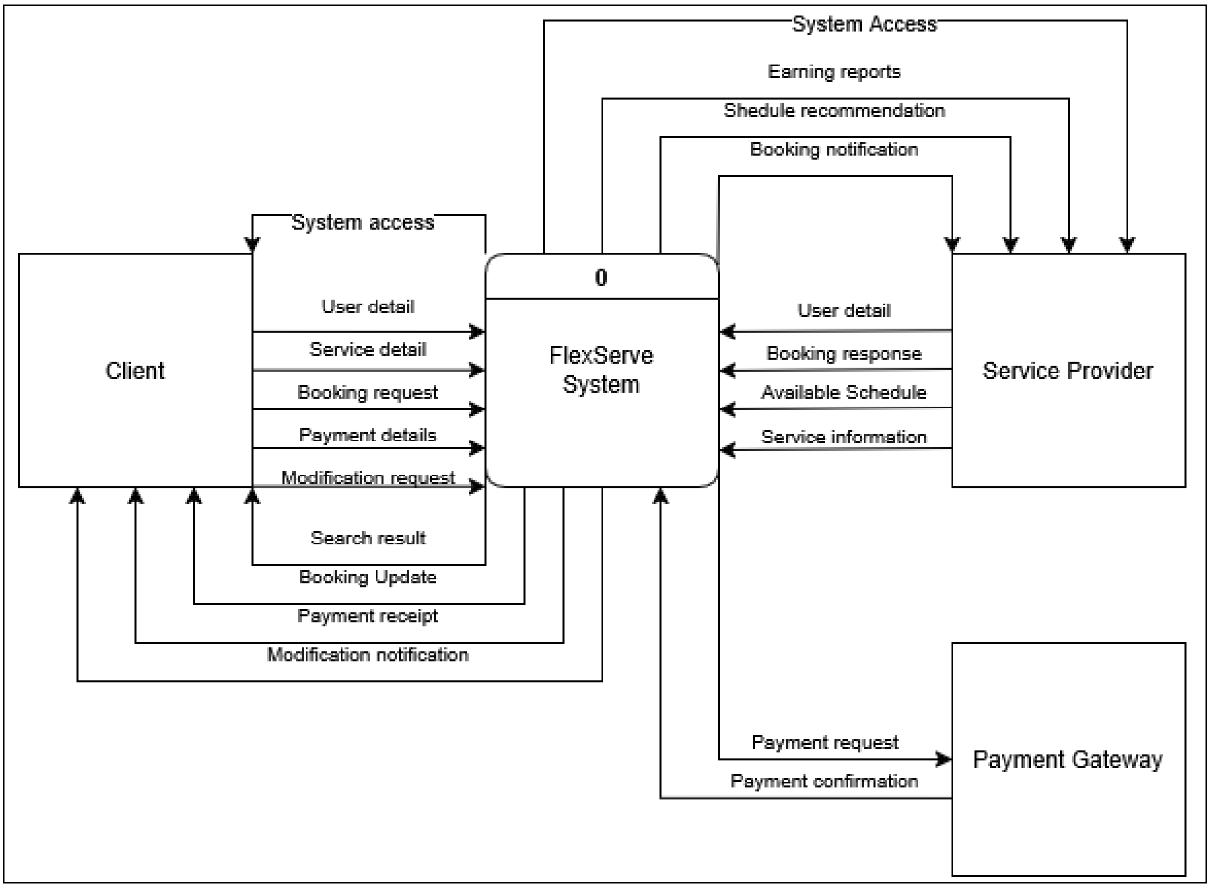

Figure 6.1.1 Context diagram of TO-BE system

Diagram 0:

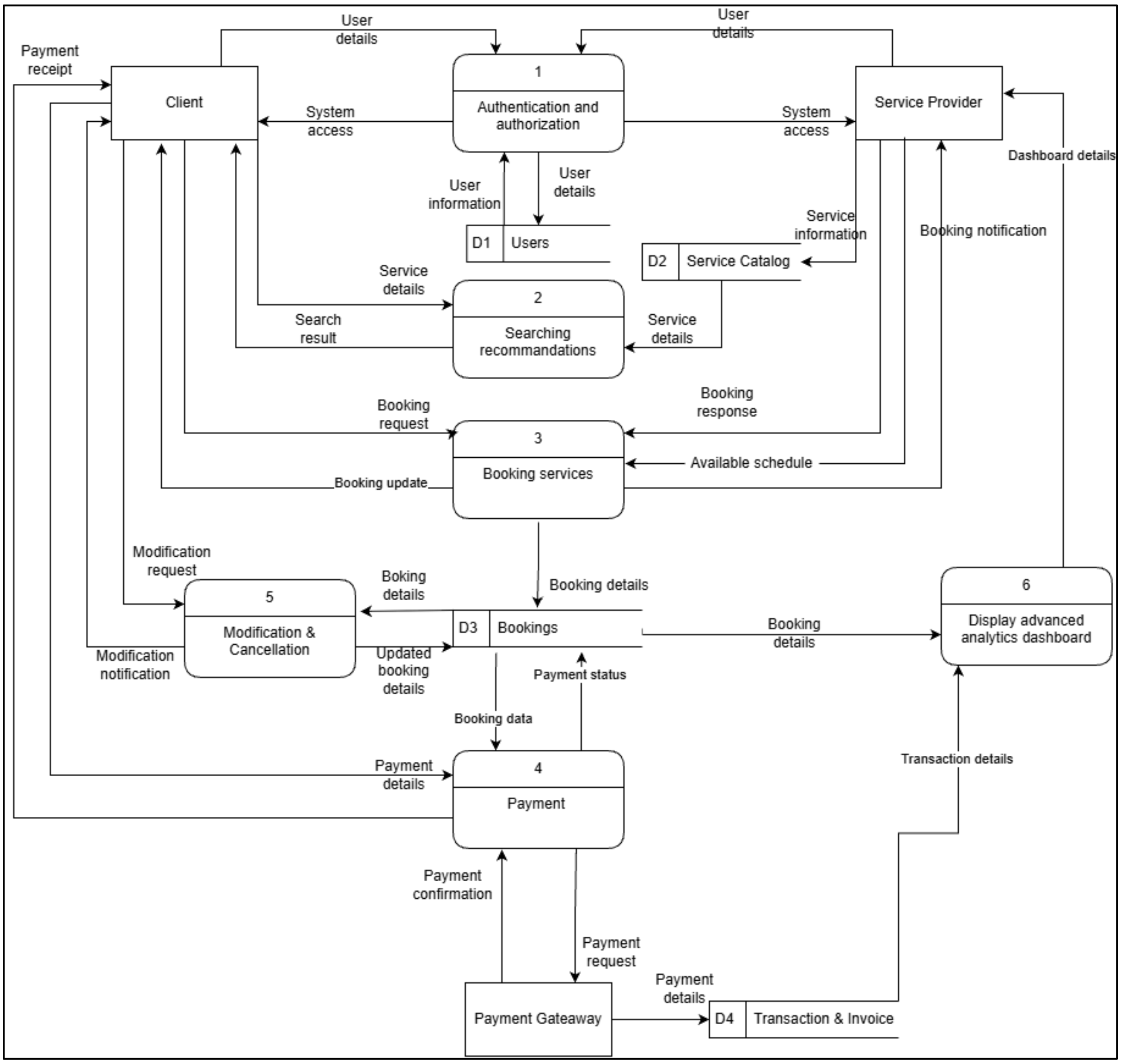

Figure 6.1.2 Diagram 0 of TO-BE system

Child diagram:

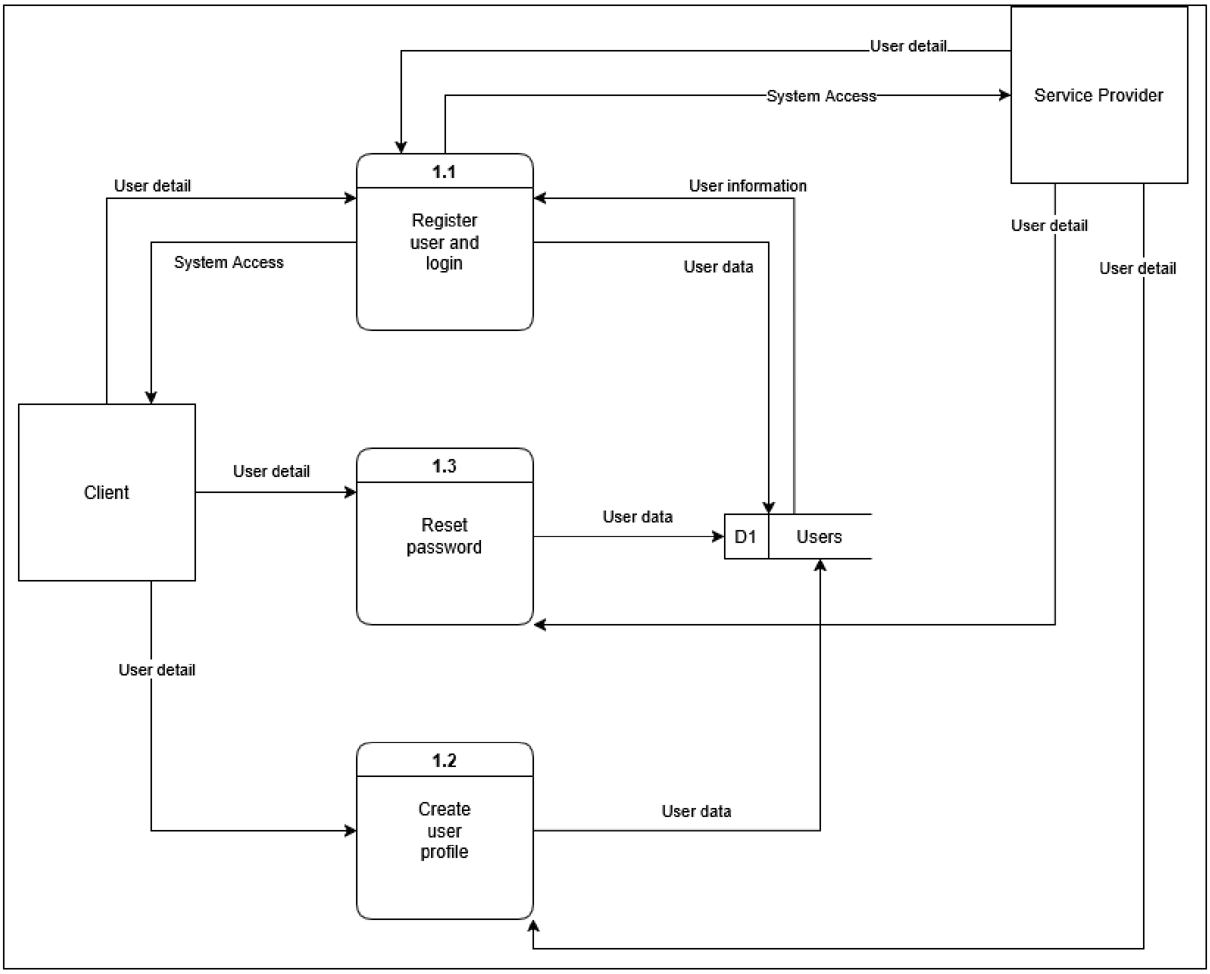

Figure 6.1.3 Child diagram of Process 1

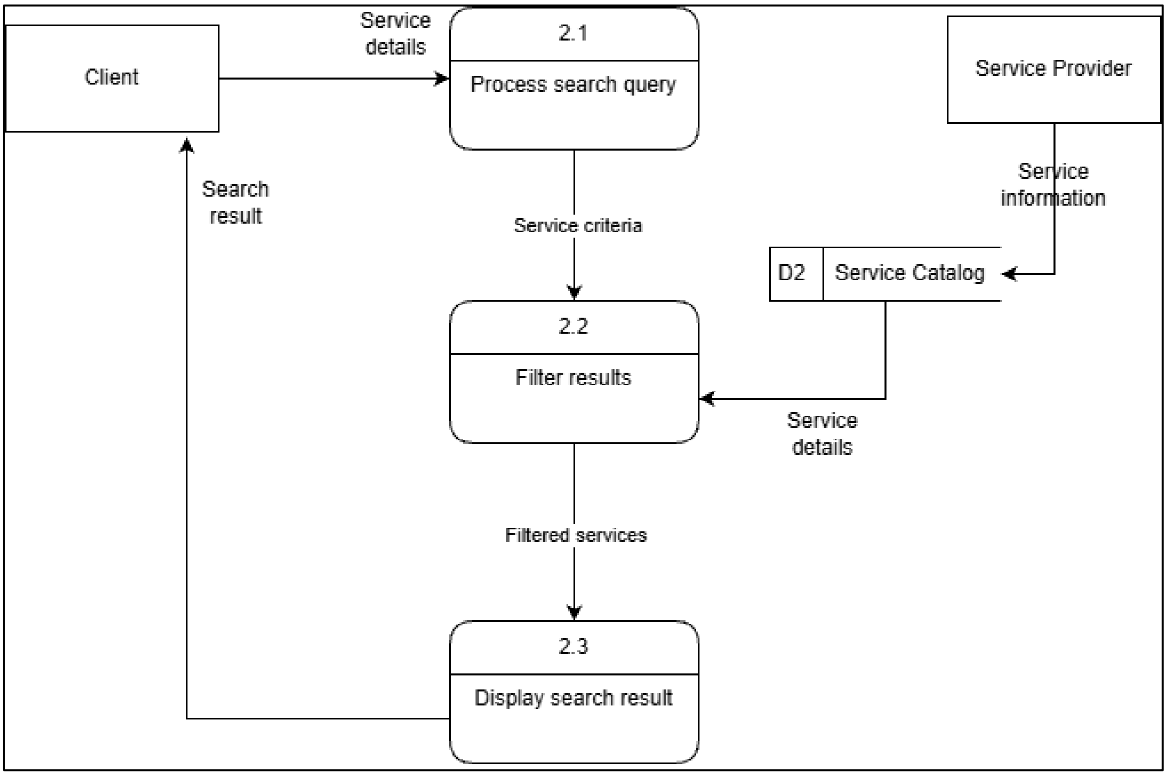

Figure 6.1.4 Child diagram of Process 2

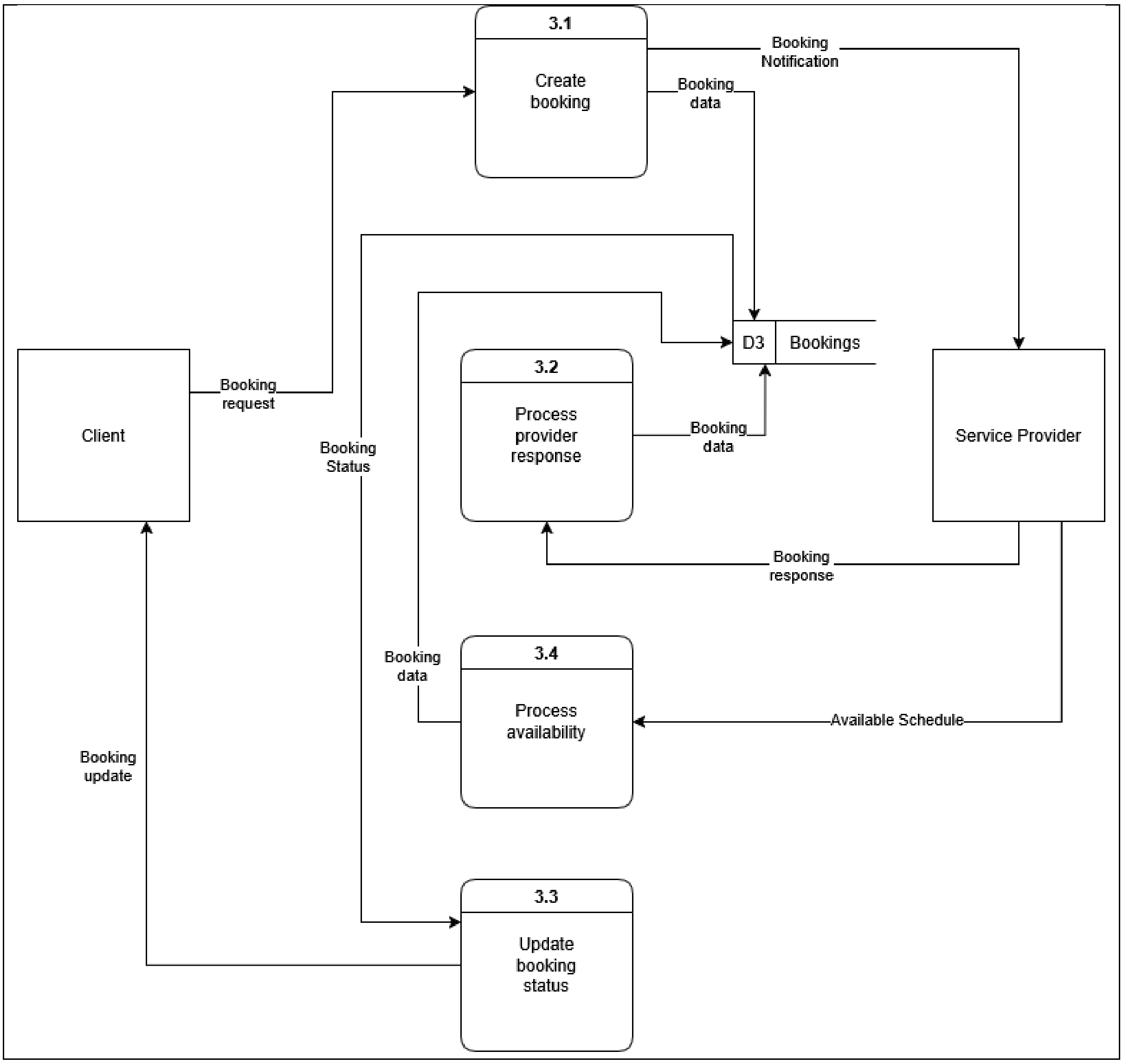

Figure 6.1.5 Child diagram of Process 3

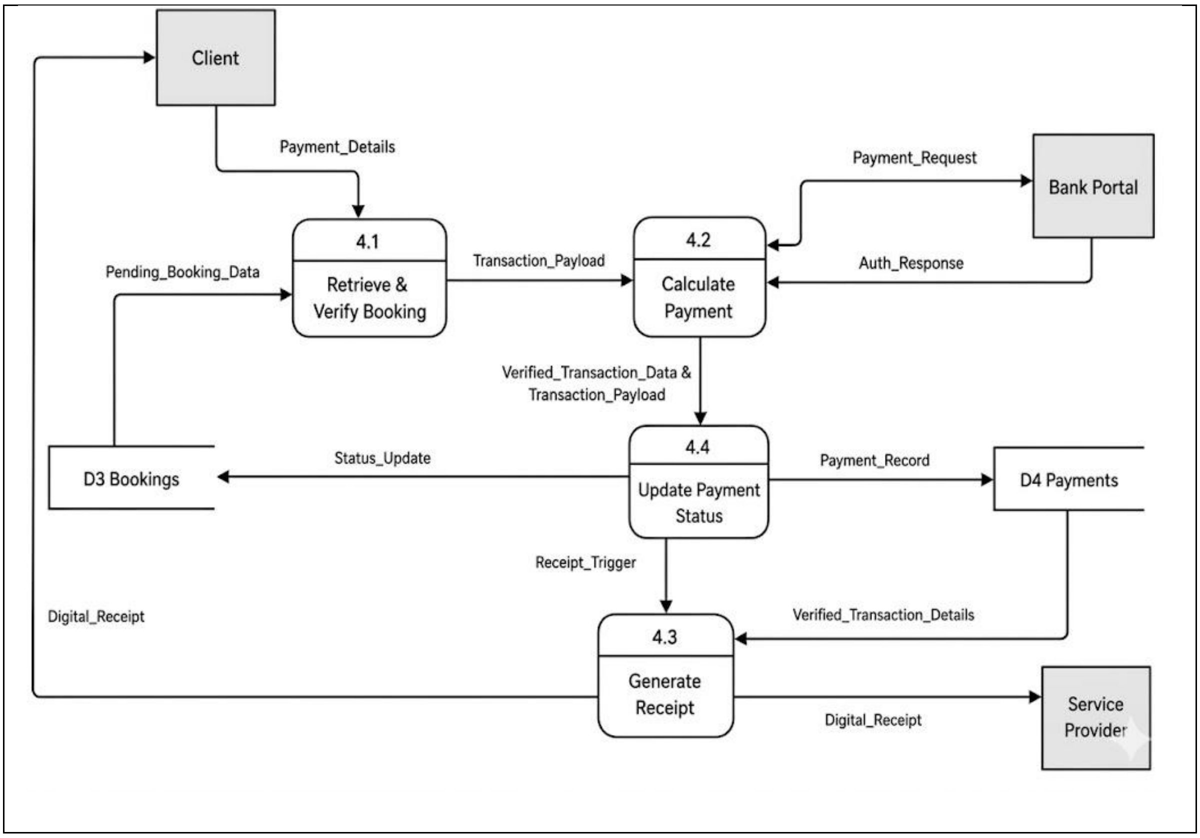

Figure 6.1.6 Child diagram of Process 4

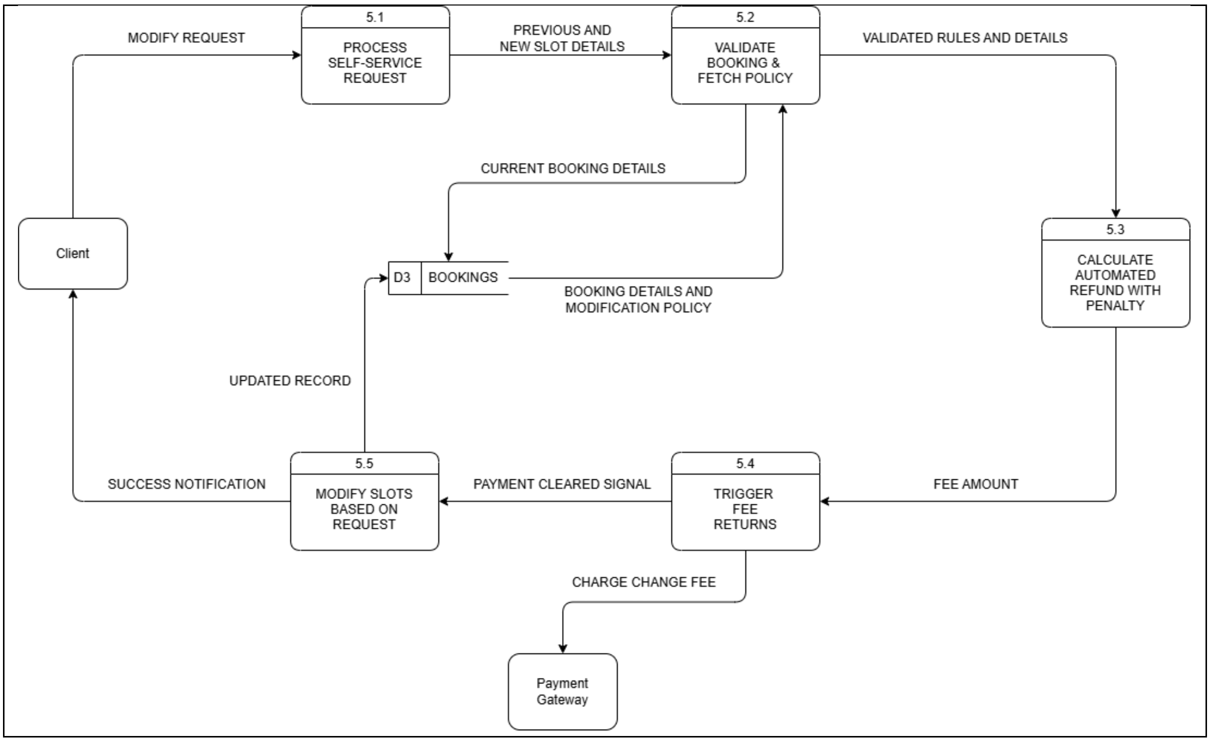

Figure 6.1.7 Child diagram of Process 5

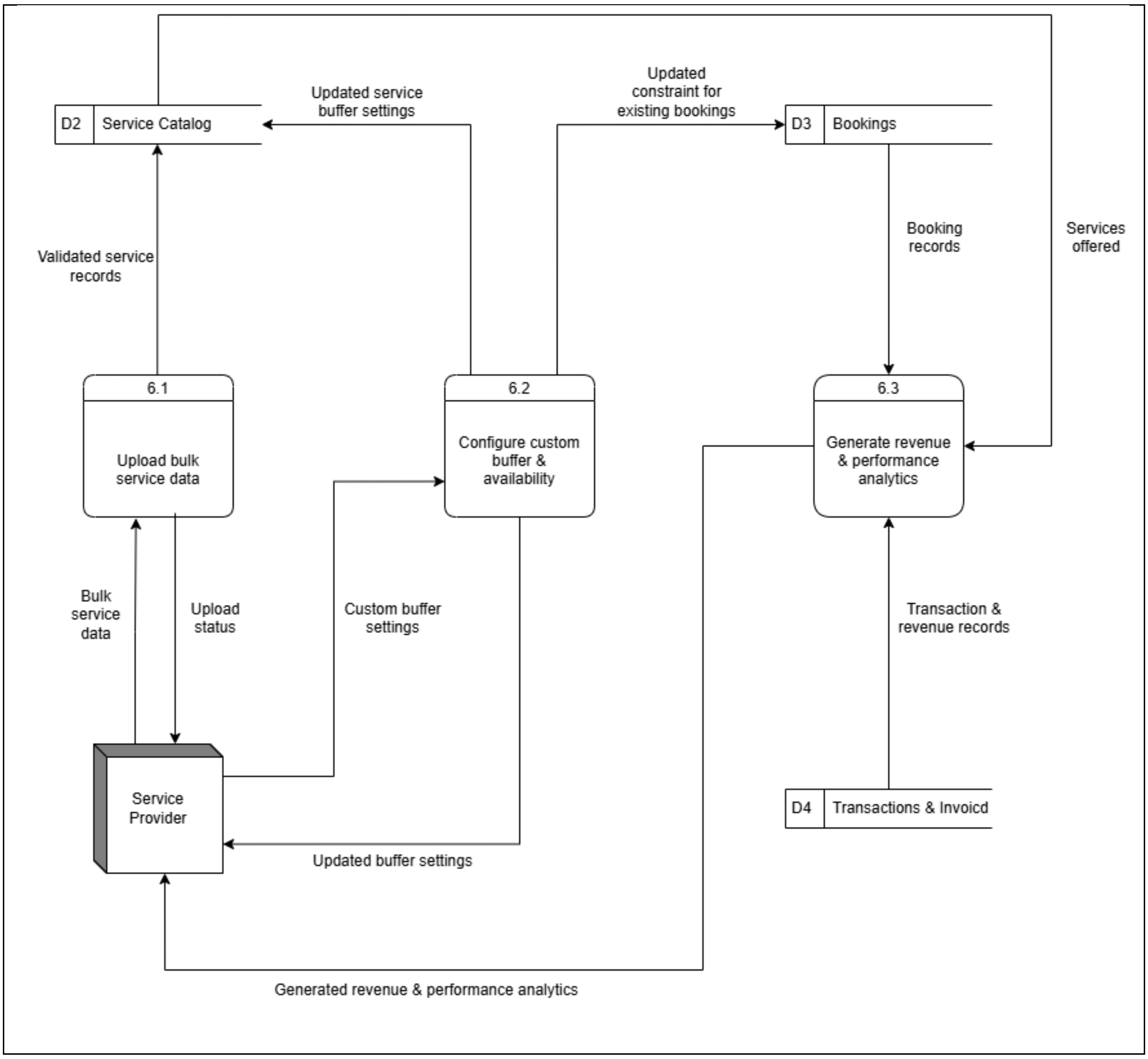

Figure 6.1.8 Child diagram of Process 6

6.2 Process Specification

```
BEGIN Manage Account
   RECEIVE user request and user detail FROM Client OR Service Provider

   IF the request is 'Register user' THEN
      CREATE new account
      WRITE new User data TO D1 Users
      SEND System Access TO Client OR Service Provider

   ELSE IF the request is 'Login' THEN
      READ User information FROM D1 Users
      COMPARE provided User detail WITH stored User information
      IF credentials match THEN
         SEND System Access TO Client OR Service Provider
      ELSE
         REJECT login attempt
      END IF

   ELSE IF the request is 'Create User Profile' THEN
      VALIDATE received User detail
      COMPILE into user data
      WRITE user data TO D1 Users

   ELSE IF the request is 'Reset Password' THEN
      READ existing User information FROM D1 Users to verify identity
      IF user identity is successfully verified THEN
         UPDATE password field in user record
         WRITE user data TO D1 Users
      ELSE
         REJECT password reset request
      END IF

   END IF
END Manage Account
```

Figure 6.2.1 Structured English of Process 1

```
BEGIN Searching Recommendations
   READ Service details from Client
   PARSE Service details into search criteria
   READ D2 Service Catalog
   FILTER Service Provider matching the search criteria
   IF filter result is empty
```

```
    DISPLAY "No Service Found" to Client
  ELSE extract the Service Provider's prices and reviews
    ADD the services details to filtered service listing
    DISPLAY the filtered service listing to Client
END IF
END Searching Recommendations
```

Figure 6.2.2 Structured English of Process 2

```
BEGIN Manage Bookings & Availability
  RECEIVE incoming booking data

  IF the incoming data is a 'Booking request' FROM Client THEN
    GENERATE booking data
    WRITE booking data TO D3 Bookings
    SEND Booking Notification TO Service Provider

  ELSE IF the incoming data is a 'Booking response' FROM Service Provider THEN

    IF Booking response is 'Accepted' THEN
      SET booking status to 'Accepted'
    ELSE IF Booking response is 'Rejected' THEN
      SET booking status to 'Rejected'
    END IF

    WRITE Booking data TO D3 Bookings
    SEND Booking update TO Client
  ELSE IF the incoming data is a 'Available schedule' FROM Service Provider THEN
    WRITE booking data TO D3 Bookings
  END IF
END Manage Bookings & Availability
```

Figure 6.2.3 Structured English of Process 3

Process 4.1: Retrieve & Verify Booking

```
BEGIN Retrieve & Verify Booking
 Retrieve Pending_Booking_Data from D3 Bookings

IF Pending_Booking_Data exists THEN
  Extract hourly_rate AND session_duration
  Extract Client_ID AND Service_Provider_ID

  Compute session_cost = hourly_rate * session_duration
```

```
   IF AI_dynamic_pricing_factor exists THEN
      Compute final_calculated_total = session_cost * AI_dynamic_pricing_factor
   ELSE
      Compute final_calculated_total = session_cost
   END IF

   Create Transaction_Payload (final_calculated_total AND
Pending_Booking_Data)

   PASS Transaction_Payload TO Process 4.2 Calculate Payment

ELSE
   RETURN "Error: Booking Record Not Found"
END IF
END Retrieve & Verify Booking
```

Process 4.2: Calculate Payment (Authorize Transaction)

```
BEGIN Calculate Payment
   RECEIVE final_calculated_total FROM Process 4.1
   RECEIVE Payment_Details FROM Client

   FORMAT Payment_Request payload (Client_ID, final_calculated_total,
Payment_Details)
   TRANSMIT Payment_Request TO Bank Portal
   WAIT FOR Auth_Response FROM Bank Portal

   IF Auth_Response is "Approved" THEN
      GENERATE Verified_Transaction_Data
      PASS Verified_Transaction_Data AND Pending_Transaction_Data TO
Process 4.4
   ELSE
      RETURN "Error: Payment Authorization Failed" TO Client
      TERMINATE Process
   END IF
END Calculate Payment
```

Process 4.4: Update Payment Status *(Note: Logically, the database must be updated before the receipt is generated to avoid printing unrecorded transactions).*

```
BEGIN Update Payment Status
   RECEIVE Verified_Transaction_Data AND Pending_Transaction_Data FROM
Process 4.2
```

```
   CREATE new payment_record IN D4 Payments using Verified_Transaction_Data
   UPDATE booking_status = "Confirmed" IN D3 Bookings for corresponding
Pending_Transaction_Data

   IF database updates are successful THEN
      TRIGGER Process 4.3
   ELSE
      LOG "Database Write Error"
      ALERT System Administrator
   END IF
END Update Payment Status
```

Process 4.3: Generate Receipt

```
BEGIN Generate Receipt
   RECEIVE Trigger FROM Process 4.4
   READ Verified_Transaction_Details FROM D4 Payments

   FORMAT Digital_Receipt using Verified_Transaction_Details
   INCLUDE date, time, Service_Provider_ID, Client_ID, and final_calculated_total

   SEND Digital_Receipt TO Client
   SEND Digital_Receipt TO Service Provider
END Generate Receipt
```

Figure 6.2.4 Structured English of Process 4

Process 5.1 Process Self-Service Request

```
BEGIN Process Self-Service Request

   RECEIVE Modify Request FROM Client
   EXTRACT Previous and New Slot Details
   SEND Previous and New Slot Details to Process 5.2

END Process Self-Service Request
```

Process 5.2 Validate Booking & Fetch Policy

```
BEGIN Validate Booking & Fetch Policy

   RECEIVE Previous and New Slot Details
   SEND Current Booking Details to D3 Bookings
   VALIDATE the Modification Request
```

FETCH Booking Details and Modification Policy from D3 Bookings
SEND Validated Rules and Details to Process 5.3

END Validate Booking & Fetch Policy

Process 5.3 Calculate Automated Refund with Penalty

BEGIN Calculate Automated Refund with Penalty

RECEIVE Validated Rules and Details
COMPUTE financial adjustments based on policy
CALCULATE Fee Amount
SEND Fee Amount to Process 5.4

END Calculate Automated Refund with Penalty

Process 5.4 Trigger Fee Returns

BEGIN Trigger Fee Returns

RECEIVE Fee Amount
SEND Charge Change Fee to Payment Gateway
WAIT for confirmation

IF payment is successful THEN
SEND Payment Cleared Signal to Process 5.5
END IF

END Trigger Fee Returns

Process 5.5 Modify Slots Based on Request

BEGIN Modify Slots Based on Request

RECEIVE Payment Cleared Signal
UPDATE booking records
SAVE Updated Record into D3 Bookings
SEND Success Notification to Client

END Modify Slots Based on Request

Figure 6.2.5 Structured English of Process 5

BEGIN Manage Service Provider Advanced Analytics Dashboard

```
   RECEIVE user request and associated data FROM Service Provider

   IF the request is 'Bulk Upload Inventory' THEN
      VALIDATE uploaded CSV/Excel file format and data integrity
      CHECK for duplicate services or missing required fields
      IF data is valid THEN
         CREATE new service records
         WRITE bulk service records TO Service Catalog
         SEND upload status success TO Service Provider
      ELSE
         SEND upload status error TO Service Provider
      END IF

   ELSE IF the request is 'Configure Custom Buffers' THEN
      READ custom buffer settings FROM Service Provider
      VALIDATE custom buffer parameters
      UPDATE service buffer rules in Service Catalog
      APPLY new availability constraints TO Bookings
      SEND configuration confirmation and impact preview TO Service Provider

   ELSE IF the request is 'View Analytics Dashboard' THEN
      READ revenue data FROM Transactions & Invoice
      READ booking performance data FROM Bookings
      CALCULATE key metrics for analytics
      GENERATE dashboard visualizations and reports
      SEND Analytics Dashboard View TO Service Provider

   ELSE IF the request is 'Export Analytics Report' THEN
      READ revenue data FROM Transactions & Invoice
      READ booking performance data FROM Bookings
      GENERATE formatted report in PDF or CSV
      SEND generated report TO Service Provider

   END IF
END Manage Service Provider Advanced Analytics Dashboard
```

Figure 6.2.6 Structured English of Process 6

7. Physical System Design

7.1 Physical DFD TO-BE system

Diagram 0

Figure 7.1.1 Physical DFD of diagram zero

Child diagram

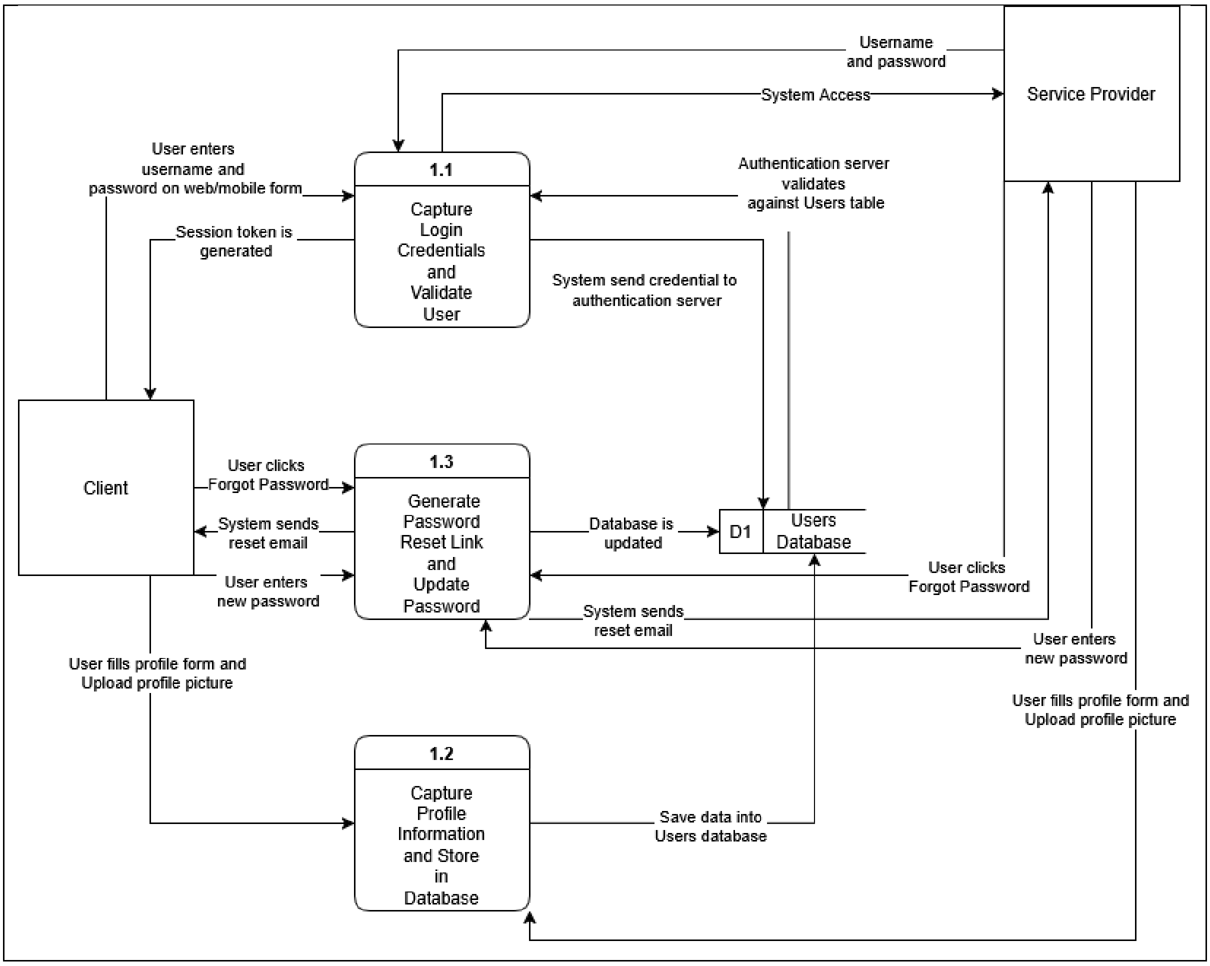

Figure 7.1.2 Physical DFD of Process 1

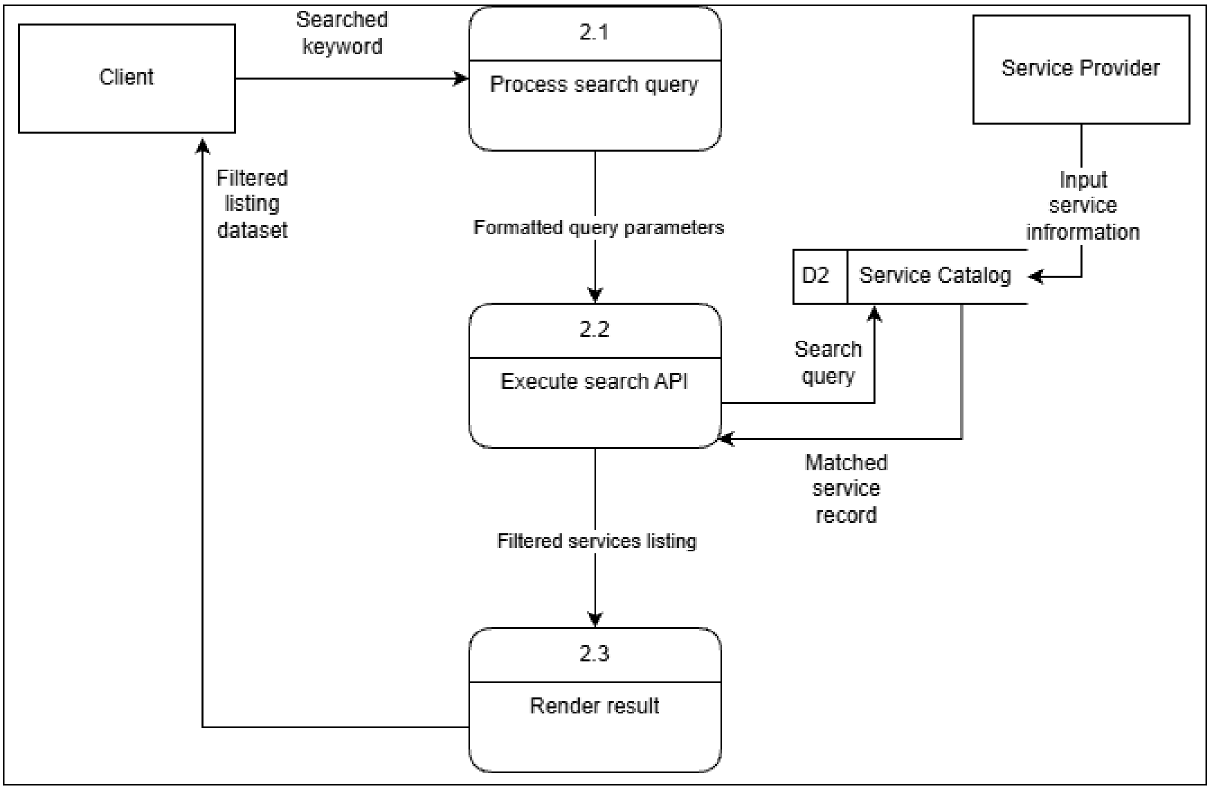

Figure 7.1.3 Physical DFD of Process 2

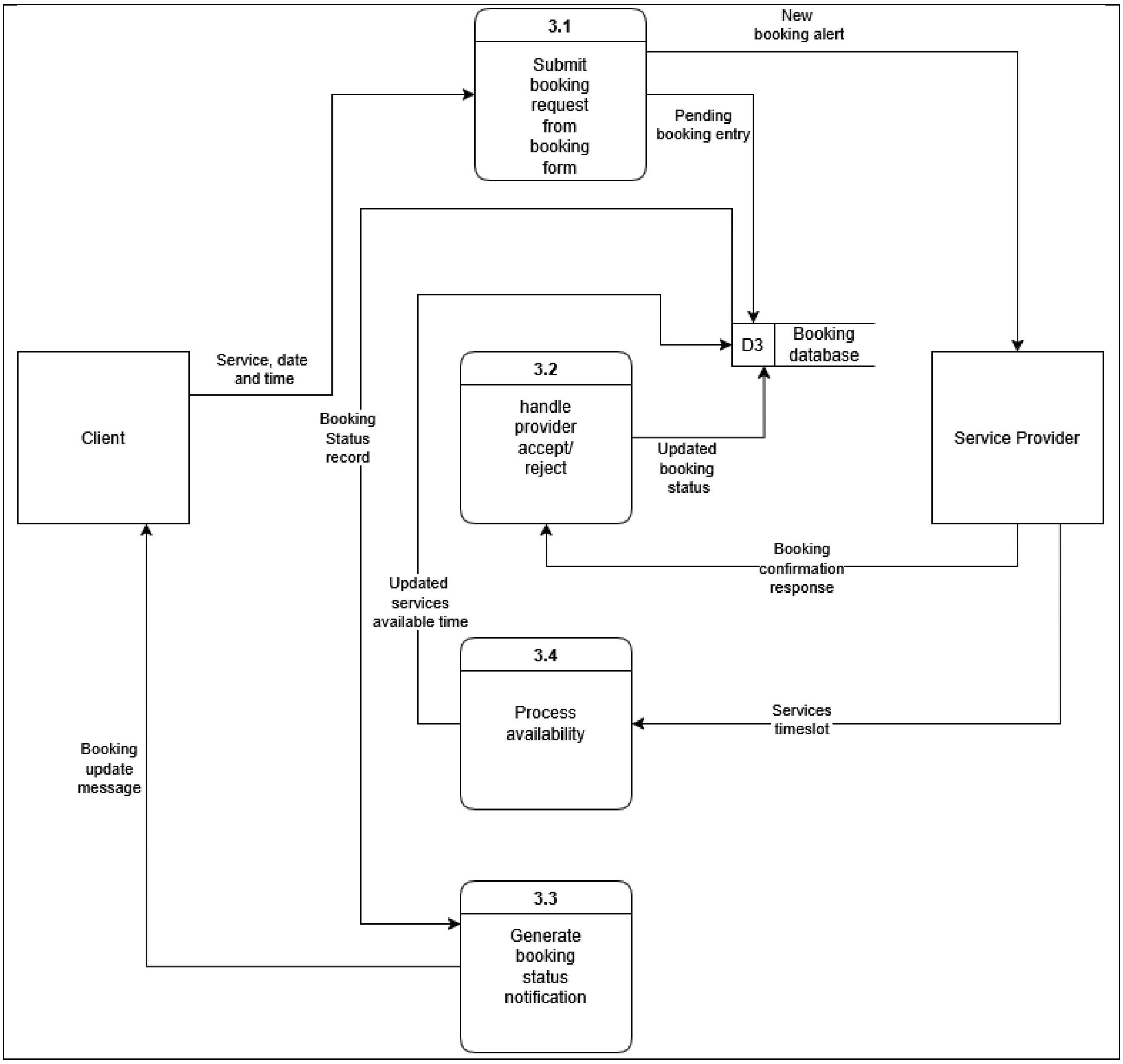

Figure 7.1.4 Physical DFD of Process 3

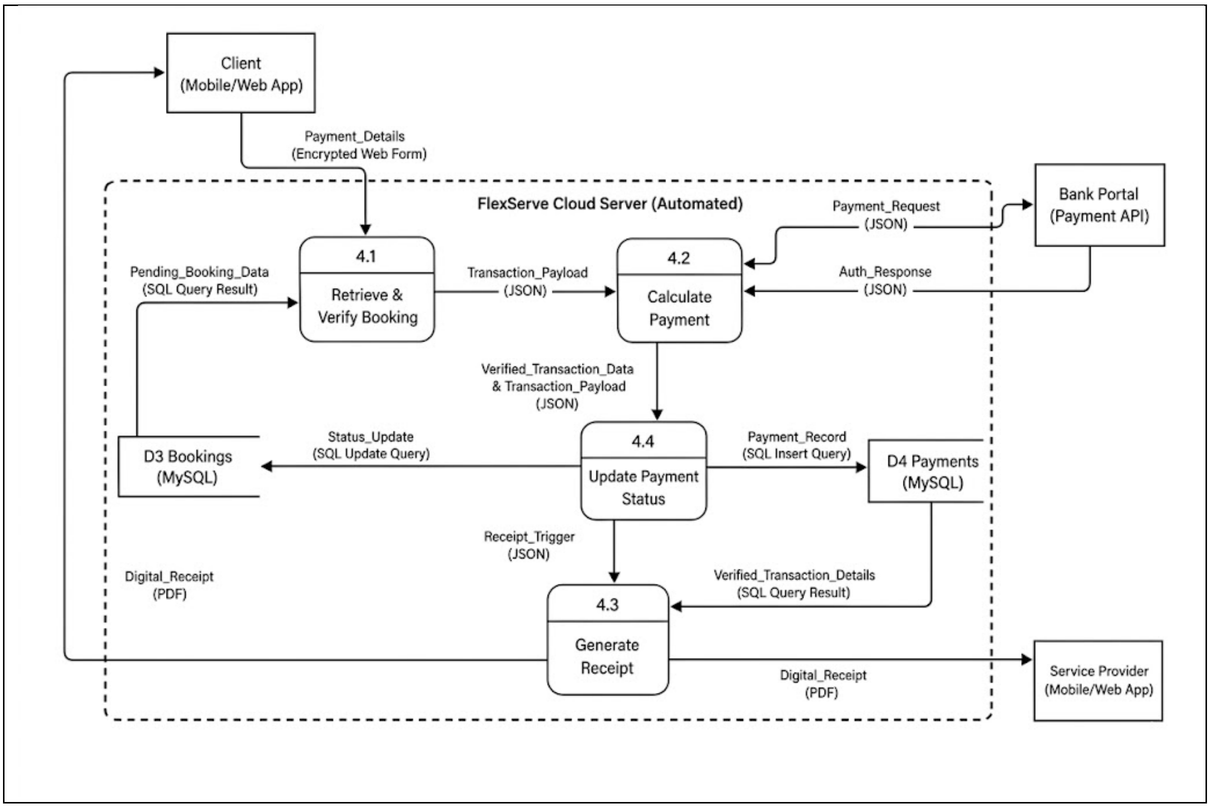

Figure 7.1.5 Physical DFD of Process 4

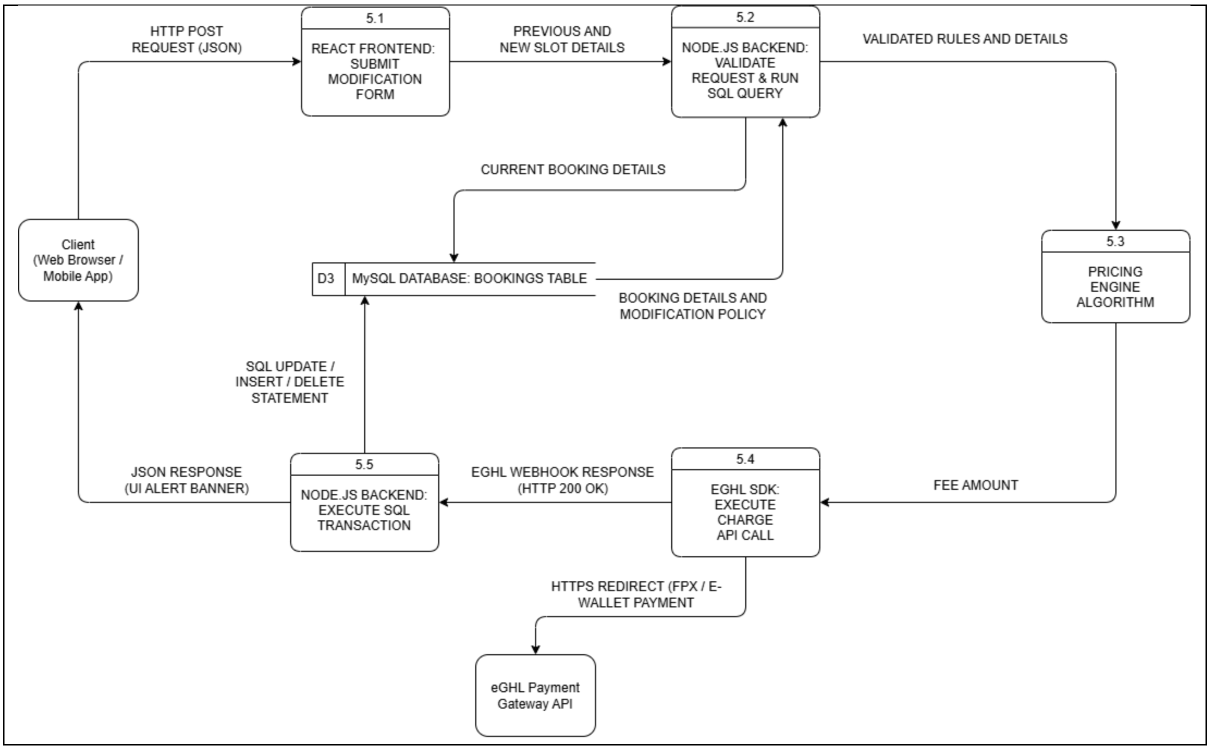

Figure 7.1.6 Physical DFD of Process 5

Figure 7.1.7 Physical DFD of Process 6

7.2 Partitioning

Figure 7.2.1 Partitioning Diagram 0

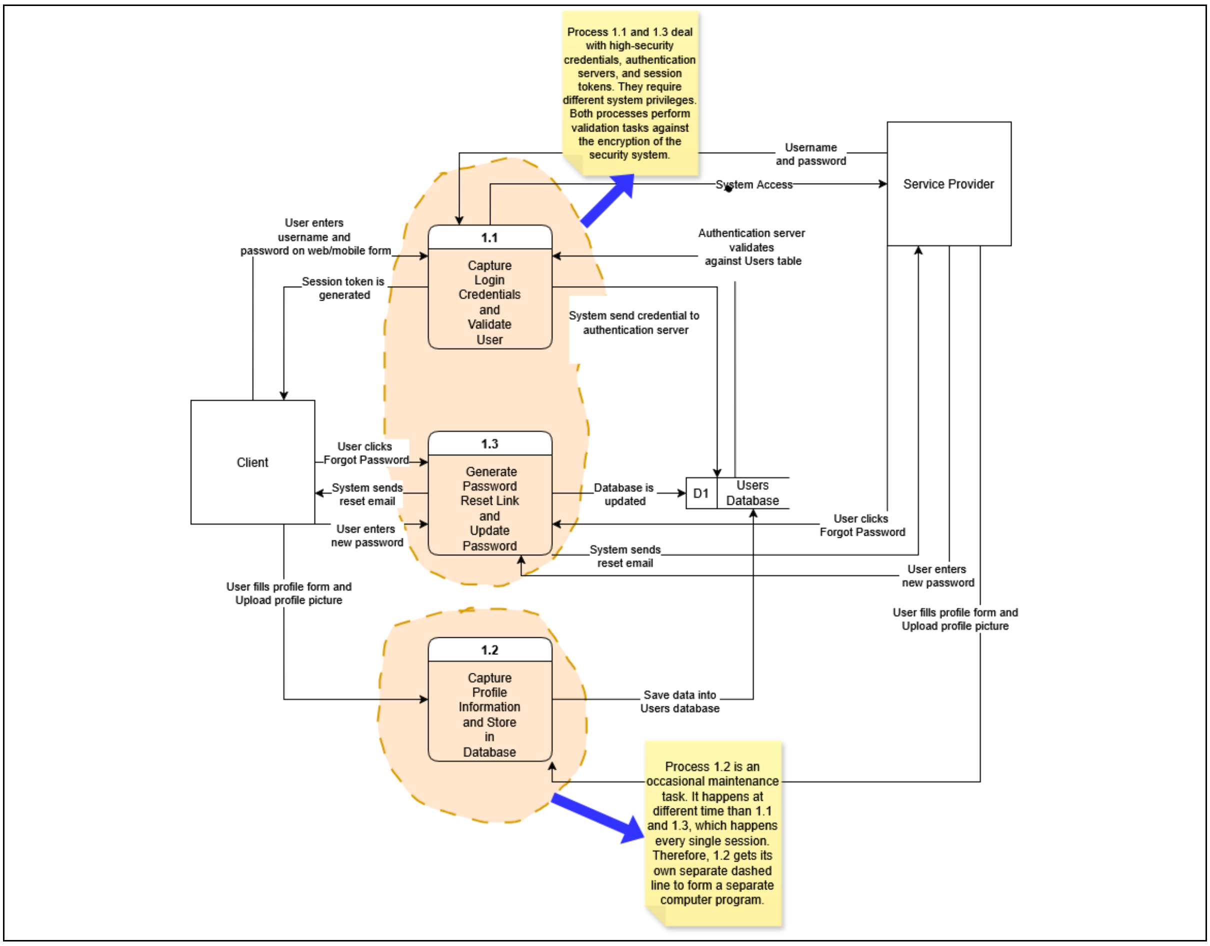

Figure 7.2.2 Partitioning Physical DFD Process 1

Figure 7.2.3 Partitioning Physical DFD Process 2

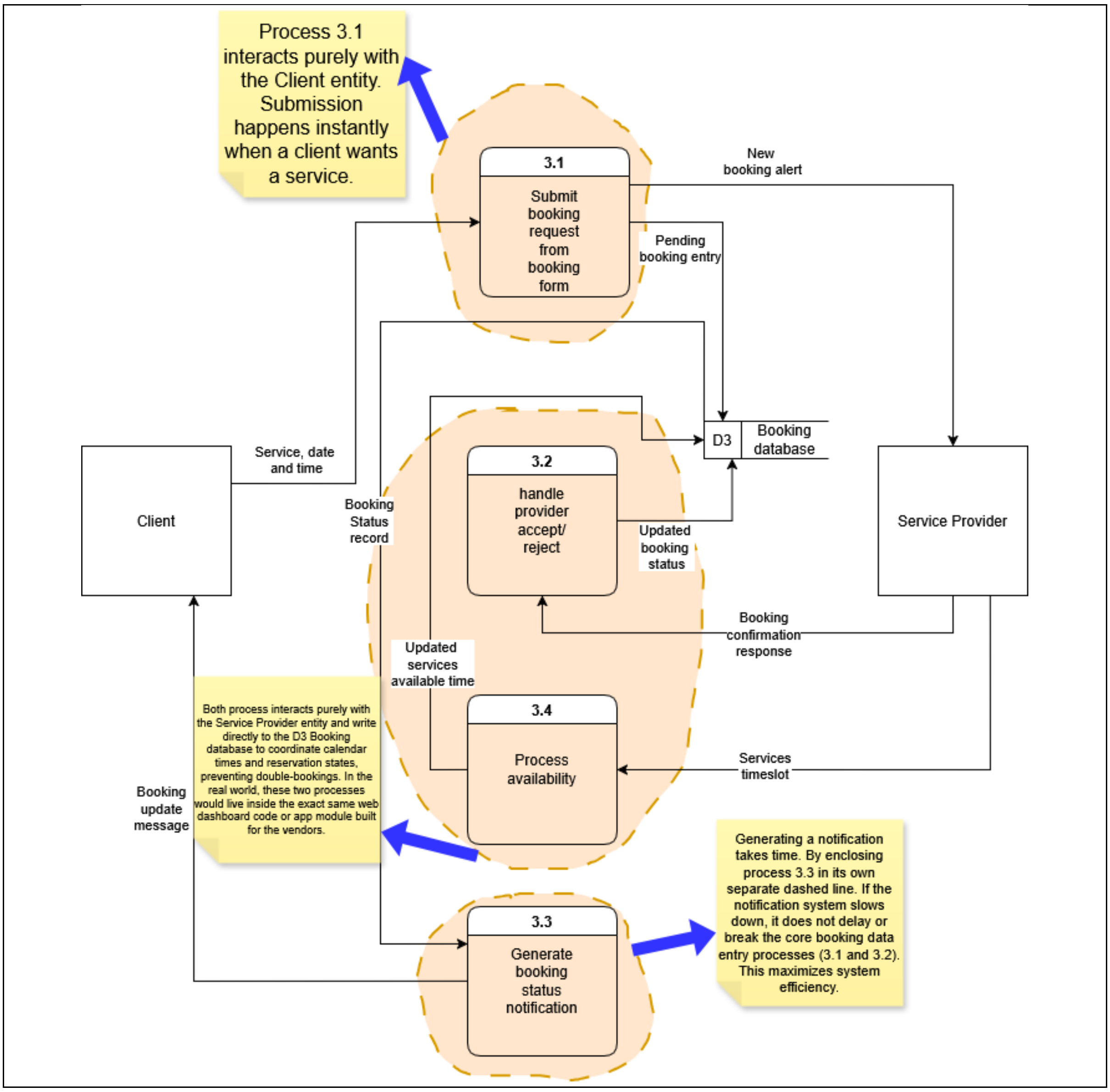

Figure 7.2.4 Partitioning Physical DFD Process 3

Figure 7.2.5 Partitioning Physical DFD Process 4

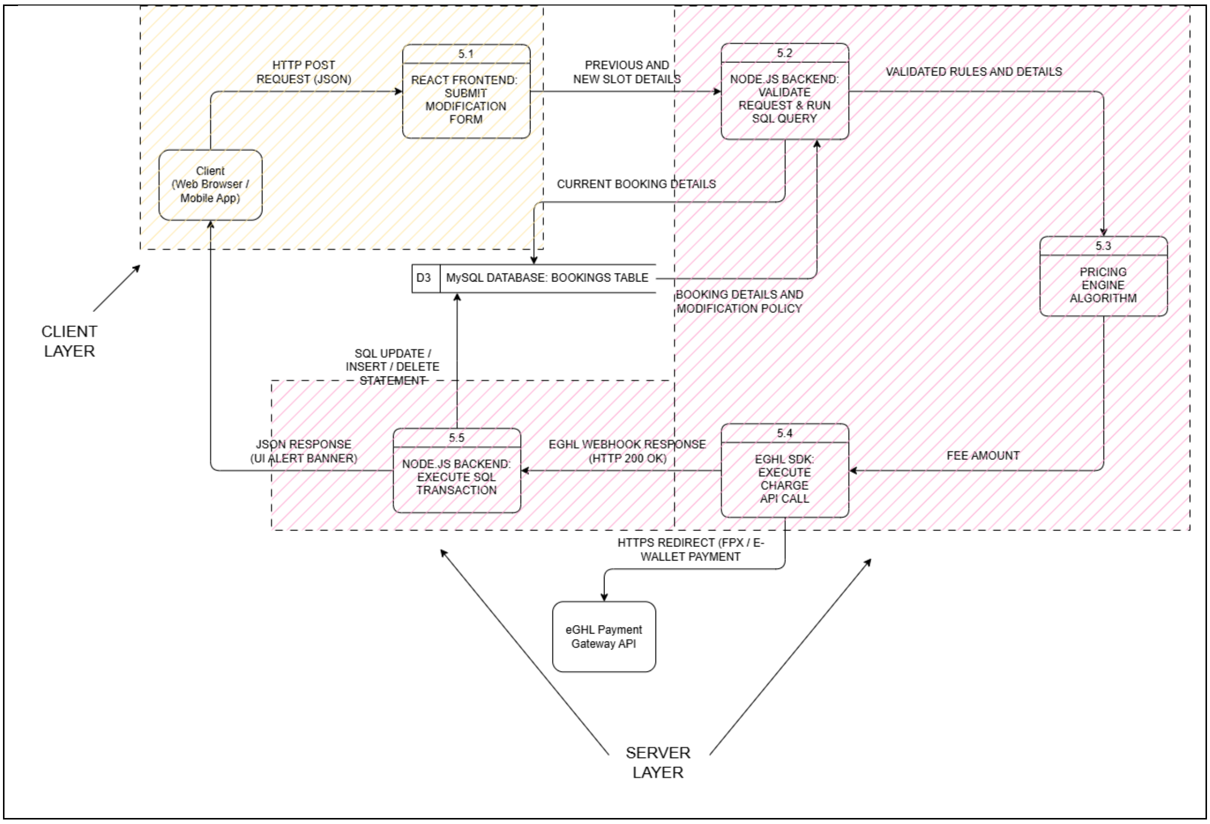

Figure 7.2.6 Partitioning Physical DFD Process 5

Figure 7.2.7 Partitioning Physical DFD Process 6

7.3 CRUD Matrix

Process Name	D1 Users
1.1 Register user and login	CR (Create, Read)
1.2 Create user profile	C (Create)
1.3 Reset password	U (Update)

Figure 7.3.1 CRUD Matrix of Process 1

Process Name	D2 Service Catalog
1.1	
1.2	
1.3	

Figure 7.3.2 CRUD Matrix of Process 2

Process Name	D3 Bookings
3.1 Create booking	C (Create)
3.2 Process provider response	U (Update)
3.3 Update booking status	R (Read)

3.4 Process availability	U (Update)

Figure 7.3.3 CRUD Matrix of Process 3

Process Name	D3 Bookings (MySQL)	D4 Payments (MySQL)
4.1 Retrieve & Verify Booking	R (Read)	–
4.2 Calculate Payment	–	–
4.3 Generate Receipt	R (Read)	R (Read)
4.4 Update Payment Status	U (Update)	C (Create)

Figure 7.3.4 CRUD Matrix of Process 4

Process Name	D3 Bookings	Client	Payment Gateway
5.1 Process Self-Service Request	-	R (Read)	-
5.2 Validate Booking & Fetch Policy	R (Read)	-	-
5.3 Calculate Automated Refund with Penalty	-	-	-
5.4 Trigger Fee Returns	-	-	U (Update)
5.5 Modify Slots Based on Request	U (Update)	U (Update)	-

Figure 7.3.5 CRUD Matrix of Process 5

Figure 7.3.6 CRUD Matrix of Process 6

7.4 Event Response Table (ERT)

Event	Source	Trigger	Activity	Response	Destination
User or Service Provider logs into the system	Client, Service Provider	User enters username and password	1.1 Capture Login Credentials and Validate User	Session token is generated, System access provided	Client, Service Provider
User or Service Provider updates profile	Client, Service Provider	User fills profile form and Upload profile picture	1.2 Create user profile	Save data into Users database	D1 Users Database
User or Service Provider requests password reset	Client, Service Provider	User clicks Forgot Password	1.3 Reset password	System sends reset email	Client, Service Provider
User or Service Provider submits new password	Client, Service Provider	User enters new password	1.3 Reset password	Update Password	Database is updated

Figure 7.4.1 ERT of Process 1

Event	Source	Trigger	Activity	Response	Destination
Client initiates a booking	Client	Service, date and time	3.1 Create booking	Pending booking entry, New booking alert	D3 Booking database, Service Provider
Service Provider updates service availability	Service Provider	Services timeslot	3.4 Process availability	Updated services available time	D3 Booking database
Service Provider confirms or rejects booking	Service Provider	Booking confirmation response	3.2 Process provider response	Updated booking status	D3 Booking database
System notifies client of booking status	D3 Booking database	Booking Status record	3.3 Update booking status	Booking update message	Client

Figure 7.4.3 ERT of Process 3

Event	Trigger Source	Use Case / Process	Response	Destination
Client initiates checkout	Payment_Details	4.2 Calculate Payment	Payment_Request	Bank Portal
Bank verifies transaction	Auth_Response	4.2 Calculate Payment	Transaction approval / rejection	System
Payment confirmed	Internal Database Trigger	4.4 Update Payment Status	Status updated in D4 Payments	D4 Payments (MySQL)
Receipt generation triggered	Receipt_Trigger	4.3 Generate Receipt	Digital_Receipt (PDF)	Client & Service Provider

Figure 7.4.4 ERT of Process 4

Event	Source	Trigger	Process	Response	Destination

Client requests for reschedule	Client	Reschedule Request	5.1, 5.2, 5.3	React Frontend sends new date by JSON, Node.js Backend checks slot availability	eGHL SDK Module
Client requests for cancellation	Client	Cancel Request	5.1, 5.2, 5.3	React Frontend sends cancel command, Node.js retrieve booking data	eGHL SDK Module
Processing transaction fee / refund	eGHL SDK	Fee amount variable passed	5.4	Executes HTTPS Redirect to handle payment	eGHL API Gateway / Client UI
Processing confirmed schedule change	EGHL Webhook	HTTP 200 OK (Paid)	5.5	Executes SQL UPDATE or DELETE command to alter slots in database.	D3 MySQL Database & Client UI

Figure 7.4.5 ERT of Process 5

7.5 Structure Chart

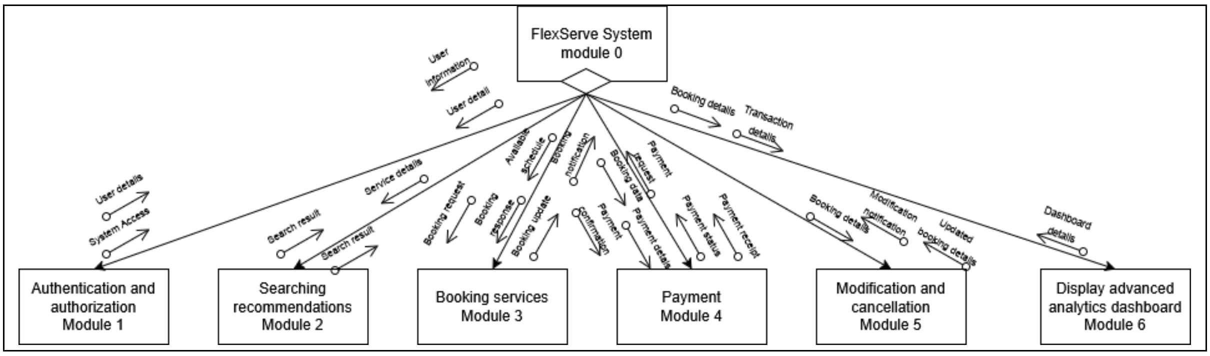

Figure 7.5.1 Structure chart of Diagram 0

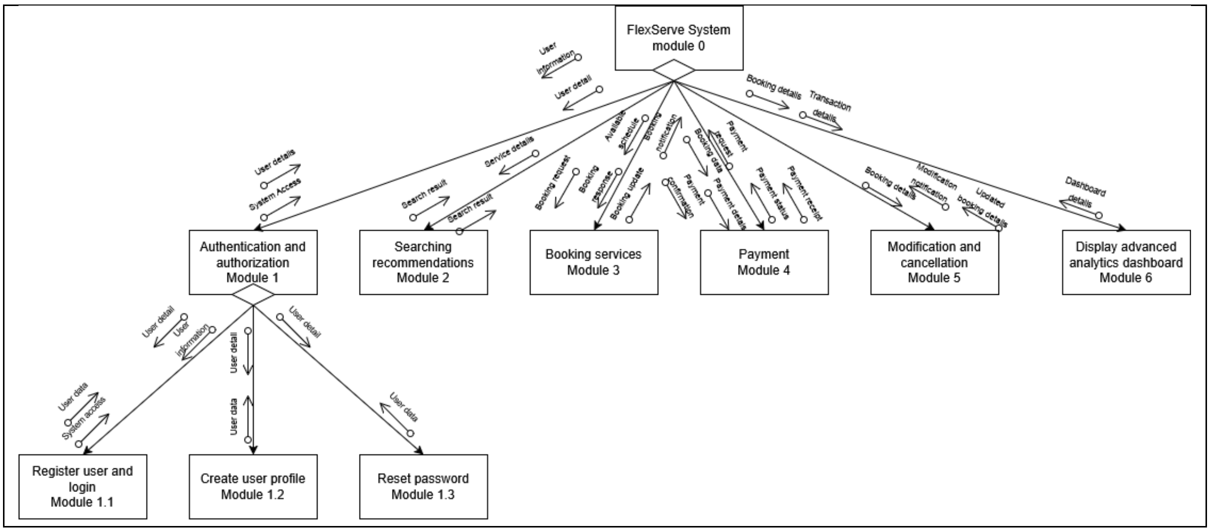

Figure 7.5.2 Structure chart of Process 1

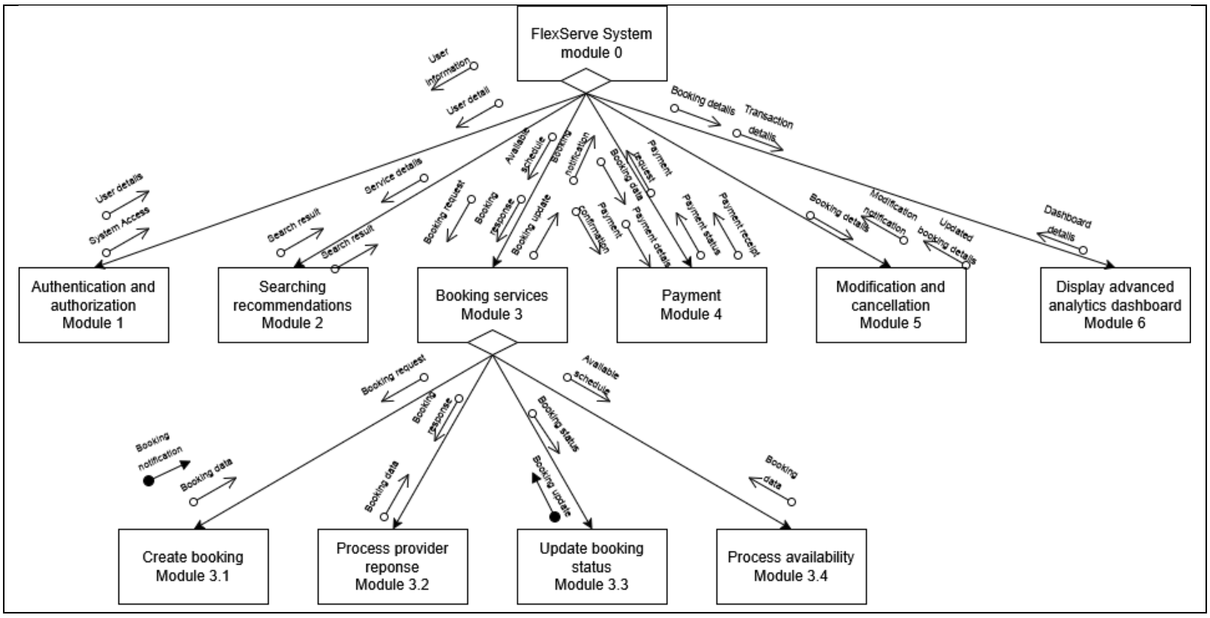

Figure 7.5.4 Structure chart of Process 3

7.6 System Architecture (fr, nfr)

8. System Wireframe

Summary of the proposed system